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ANTHROPOLOGY

Paleolakes and socioecological implications of last glacial "greening" of the South African interior

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Determining the timing and drivers of Pleistocene hydrological change in the interior of South Africa is critical for testing hypotheses regarding the presence, dynamics, and resilience of human populations. Combining geological data and physically based distributed hydrological modeling, we demonstrate the presence of large paleolakes in South Africa's central interior during the last glacial period, and infer a regional-scale invigoration of hydrological networks, particularly during marine isotope stages 3 and 2, most notably 55 to 39 ka and 34 to 31 ka. The resulting hydrological reconstructions further permit investigation of regional floral and fauna responses using a modern analog approach. These suggest that the climate change required to sustain these water bodies would have replaced xeric shrubland with more productive, eutrophic grassland or higher grass-cover vegetation, capable of supporting a substantial increase in ungulate diversity and biomass. The existence of such resource-rich landscapes for protracted phases within the last glacial period likely exerted a recurrent draw on human societies, evidenced by extensive pan-side artifact assemblages. Thus, rather than representing a perennially uninhabited hinterland, the central interior's underrepresentation in late Pleistocene archeological narratives likely reflects taphonomic biases stemming from a dearth of rockshelters and regional geomorphic controls. These findings suggest that South Africa's central interior experienced greater climatic, ecological, and cultural dynamism than previously appreciated and potential to host human populations whose archaeological signatures deserve systematic investigation.

Pleistocene | southern Africa | paleolake | middle stone age

Significance

Renowned for its rich evidence pertaining to early *Homo sapiens*, South Africa's late Pleistocene archaeological record remains biased toward the continental margins. A profusion of coastal rockshelters, together with putative associations of glacial periods with intense aridity in the continental interior, has created historical impressions of an inland core hostile to human life during lengthy glacial phases. New geological data and large-scale hydrological modeling experiments reveal that a series of large, now-dry palaeolakes existed for protracted phases of marine isotope stages 3 and 2. Modern analog reconstructions of resulting vegetational and faunal communities suggest that glacial climates were capable of transforming South Africa's central interior into a resource-rich landscape favorable to human populations.

South Africa's archaeological record is world renowned for a diversity of evidence relating to early *Homo sapien* lifeways (1). This record is biased toward the continental margins (Fig. 1), due partly to the favorable formation and preservation of rockshelters in such locales. Combined with historical assumptions concerning aridity-driven inhospitality of the continental interior, this has contributed to an emphasis on the significance of coastal regions for early modern human populations (e.g., refs. 2–9), with the southern Cape's now-submerged coastal platform posited as a glacial-phase refugium (6–8). *Sensu lato*, southern Africa's currently semi-arid interior basins are climatically and ecologically complex. Portions of them clearly experienced conditions amenable to human settlement at various times during the middle and late Pleistocene, as demonstrated by recent work in, for example, the northern and southern Kalahari (10–13), the Northern Cape (14), the Tankwa Karoo (15), and the eastern Free State (16). However, the precise timing and nature of environmental changes in South Africa's central interior (Fig. 1)—the semi-arid zone nearest the rich and well-studied southern Cape archaeological records—remain largely obscure, as does the nature of the attendant resources that drew hunter-gatherers to it. Given these uncertainties, the effects of middle and late Pleistocene climate variability on human migration, dispersal, and adaptation—in southern Africa as elsewhere—are subjects of ongoing debate (17, 18).

Key aspects of this debate can be informed by an improved understanding of the evolution of surface water systems (10, 19). Specifically, endoreic paleolake systems are sensitive indicators of precipitation–evaporation balance (e.g., refs. 20 and 21), and thus particularly amenable to integration with hydrological and paleoclimatic models (e.g., ref. 22). In the central interior of South Africa, such paleolake basins (pans and playas)—often with rich assemblages of Middle and Later Stone Age (MSA and LSA) archaeological materials—have been described (14, 23–25). However, their geochronologies and climatic significance remain very poorly constrained. To redress this, we present a suite of luminescence ages (n = 23) and radiocarbon dates (n = 18), which together with field mapping, provides fresh perspectives on the timings of lacustrine phases in the central interior. We evaluate the significance of these data using a physically based distributed hydrological model applied to the wider Orange River basin, which enables the reconstruction of regional paleohydrological and

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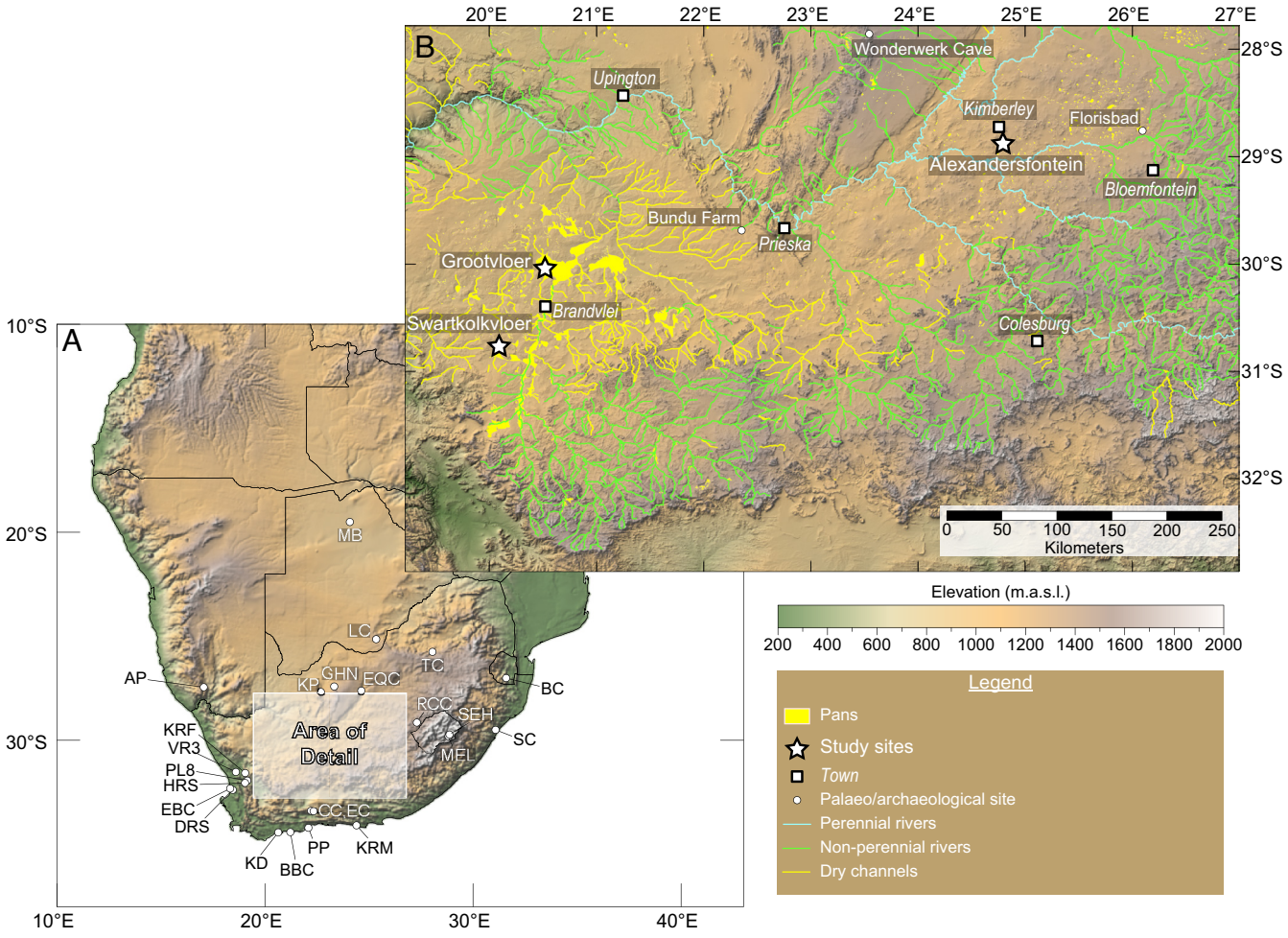


Fig. 1. Map of South African study region with modern hydrologic system, study sites, and relevant paleoenvironmental and archaeological sites. Site codes indicated are as follows: AP = Apollo 11 Rockshelter; BBC = Blombos Cave; BC = Border Cave; CC = Cango Cave; DK = Die Kelders Cave; DRS = Diepkloof Rockshelter; EBC = Elands Bay Cave; EC = Efflux Cave; EQC = Equus Cave; GHN = Ga Mohana Rockshelter; HRS = Hollow Rock Shelter; KD = Klipdrift Cave; KFR = Klipfonteinrand; KP = Kathu Pan; KRM = Klasies River Mouth; Lobatse Cave = LC; MB = Makgadikgadi Basin; MEL = Melikane Rockshelter; PP = Pinnacle Point sites; PL8 = Putslaagte 8; RCC = Rose Cottage Cave; SC = Sibudu Cave; SEH = Sehonghong Rockshelter; TC = Tswaing Crater; VR3 = Varsche River 3.

paleoecological conditions. The results indicate that during the last glacial period, for extended periods of (marine isotope stages) MIS 3 in particular, the South African central interior was a potentially resource-rich landscape that would have been attractive to Pleistocene hunter-gatherer populations.

Results and Discussion

For this study, we investigated several pans across the South African interior (Fig. 1). The most extensive suite of unequivocal lacustrine deposits is preserved at Swartkolkvloer, a large pan (~70 km^2, 921 m a.s.l.) south of Brandvlei (Figs. 1 and 2, *SI Appendix*, p. 9). Shorelines—at 11 m and 18 m above the modern pan floor—are located on the pan's southern and eastern margin at Breek Been Kolk (BBK; (14)) and "Locality A" (23), while lagoonal sediments at +13 m were observed at the "Quarry." These deposits comprise a mixture of lacustrine marls, abundant shells of the aquatic gastropod *Tomichia ventricosa,* and water-rolled gravels (23). Together, they indicate that a ~215 km^2 lake was present for sustained periods (Fig. 2). Luminescence and radiocarbon dating (n = 19) indicate lacustrine phases from ~39 ± 3 to 55 ± 3 ka (939 m, +18 m) and from 31.4 ± 0.3 to 34.4 ± 0.7 cal kBP (932 m, +11 m), with a later, perhaps more ephemeral, phase potentially indicated by OSL-dated lacustrine sediments (17 ± 2 ka) and by previously reported *T. ventricosa* radiocarbon dates at BBK (14) (Fig. 2, *SI Appendix*, p. 12). Broadly consistent with these ages is the observation of a MSA noncentripetal lineal Levallois core on the 932 m Locality A shoreline, as well as MSA artifacts at adjacent Hoek van Spruit se Vloer. The latter matches descriptions of Sampson's (26) later Orangian Industry, dated to MSA 3 at Driekoppen (27) (*SI Appendix*, pp. 22, 25). Stable isotope analyses of *T. ventricosa* shells (*SI Appendix*, p. 26) reveal strong $\delta^{13}C$ and $\delta^{18}O$ covariance, consistent with a closed lake basin environment (28). The lowest $\delta^{13}C$ and $\delta^{18}O$ values, indicative of the least evaporative conditions, were observed for specimens from the +18 m lake level at BBK.

Approximately 100 km north of Swartkolkvloer, *Unio caffer* shells from pan fill sediments at Grootvloer were radiocarbon dated to 20.7 ± 0.2 and 21.7 ± 0.6 cal kBP (*SI Appendix*). *U. caffer,* absent from this region today, requires perennial freshwater and the presence of fish to reproduce (36). Subjacent pan fill produced similar luminescence ages across three locations (20 ± 1 to 24 ± 2 ka; n = 3; *SI Appendix*, p. 35), indicating a lacustrine phase during the Last Glacial Maximum. A pan-margin artifact scatter rich in high-quality bladelets, bladelet and bipolar cores, and weathered ostrich eggshell provides evidence of a contemporary (early Robberg) human presence (*SI Appendix*, p. 37).

Five hundred kilometers to the northeast at Alexandersfontein, lacustrine beds 14.5 m above the modern pan floor (see ref. 24)

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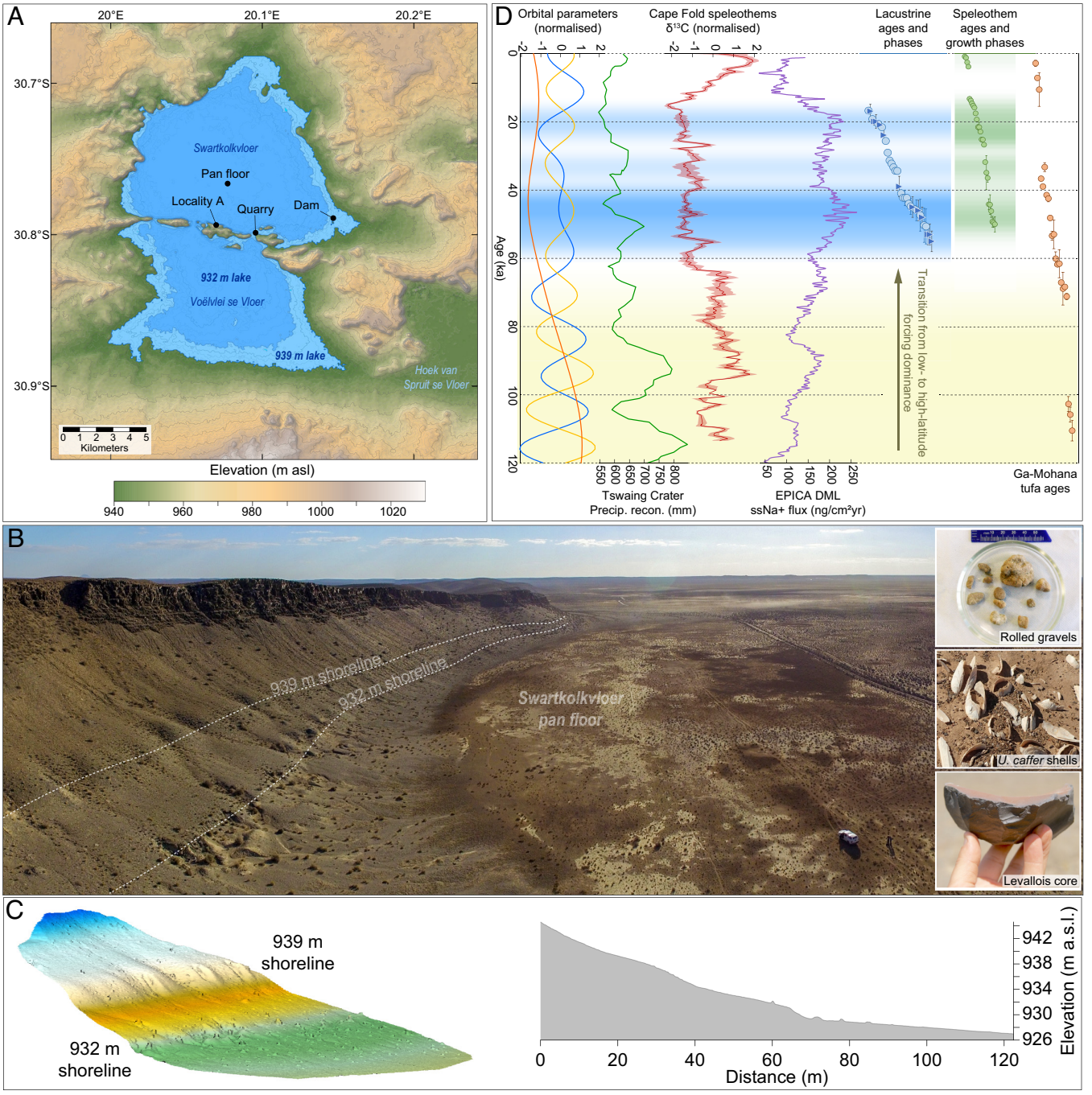


Fig. 2. Map of Swartkolkvloer (*A*) with the locations of the sampling sites. View from the east (*B*) shows the upper and lower shorelines [also rendered in a photogrammetric model and derived slope profile (*C*)]. Example of evidence for sustained lacustrine conditions in the form of rolled gravel and freshwater molluscs are shown to the right of panel *C*, as is a Levallois core found along the lower shoreline. Panel *D* provides a comparison of optically stimulated luminescence (triangles) and ^{14}C (circles) ages obtained from lacustrine materials as part of this study from the South African interior (blue) with orbital parameters (29), the Tswaing Crater precipitation reconstruction (30), speleothem $\delta^{13}C$ data from the Cape Fold mountains (31, 32) indicating the transition to cooler glacial conditions, the EPICA DML sea-salt sodium flux as a proxy for Antarctic sea-ice (33), and interpolated speleothem growth ages from Wonderwerk (34) and Lobatse (35) caves. Yellow shading indicates transition from low-to-high latitude forcing dominance as inferred from regional data (31).

were radiocarbon dated to between 25.7 ± 0.7 and 32.3 ± 0.7 cal kBP. The minimum age of these beds is constrained by early Holocene luminescence ages from the overlying colluvium (9 to 10 ka). MSA lithic artifacts occur as a discontinuous, low-density scatter on the surfaces of the exposed lacustrine beds and at the base of the gullies formed within these beds. We estimate a ~35 km^2 lake associated with this water depth (14.5 m), compared with the ~44 km^2 19 m deep lake previously inferred to have developed during the LGM at 19.3 ± 0.7 cal kBP (24). Evidence for such a 19 m high-stand could not be (re)located (see *SI Appendix* discussion p. 39). At a second site, Mauritzfontein, on the southern margin of the Alexandersfontein basin and ~10 m above the modern pan floor, late MIS 2 (13 ± 1 ka) lacustrine marls overlie evidence for a lower MIS 3-2 lake phase (tentatively ~41 ka; *SI Appendix*, p. 46), the latter comprising a wave-cut calcrete and beach sands with in-situ MSA artifacts (24, 37).

To contextualize these data, we used the Shetran physically based distributed hydrological model (38) to: 1) determine the

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conditions necessary to establish and maintain lake high-stands at Swarkolkvloer and Alexandersfontein, and 2) to estimate the resulting regional-scale hydrological rejuvenation of the wider Orange and Vaal river basins (Fig. 1). While we are not able to consider the impact of changes in the magnitude and frequency of rainfall events on water balance during the study period, results nonetheless indicate that to sustain the observed lacustrine features, increases in precipitation (P) to potential evapotranspiration (PET) ratios of 75% (+18 m lake at Swarkolkvloer) and 60% (+19 m lake at Alexandersfontein), are required (*SI Appendix*, Table S9). Applying paleotemperature estimates for the lacustrine phases (*Materials and Methods*), we calculate that PET at Swarkolkvloer and Alexandersfontein was 21% and 25% lower than present, respectively, and precipitation increased by ~215% at Swarkolkvloer (to 550 mm/y) and 88% at Alexandersfontein (to 820 mm/y), conditions that would have led to a widespread rejuvenation of hydrologic systems in the central interior of South Africa (*SI Appendix*, p. 56).

From these data, we therefore infer that substantial humidity increases were required to sustain perennial surface water in the South African central interior during MIS 3-2 (Figs. 2 and 3). The regional significance of these observations is apparent when considered in broader palaeoclimatic context, and in light of evidence of increased MIS 3 humidity and hydrologic activity at other locales in the interior (e.g., refs. 12, 34, 35, 39, and 40). These high lake phases postdate the region's eccentricity-modulated transition from low to high latitude forcing dominance at ~70 ka (31) and correspond to periods of Antarctic sea-ice expansion from ~58 to 16 ka (33; Fig. 2). The highest shoreline at Swarkolkvloer is coincident with the precession-induced sea-ice maximum from ~55 to 40 ka, implying that equatorward displacements of the westerly storm track and increased winter rainfall may have been instrumental in establishing sustained lacustrine phases. This mechanism may have operated in isolation, or as part of a more complex dynamic, wherein frontal systems were brought into more direct association with tropical lows—favored by increased austral summer insolation—to form composite synoptic systems (i.e., tropical–temperate troughs) (41, 42). Indeed, stable carbon isotope evidence from the tooth enamel of grazing ungulates at Equus Cave indicates a moderate increase in cool-growing season C_3 grasses during late MIS 3 and MIS 2 (43), implying that while winter rainfall likely increased, it did not dominate the regional rainfall regime. Phases of speleothem growth at nearby Wonderwerk (34) and Lobatse (44) caves (50 to 44 ka, 31 to 36 ka and 26 to 13 ka) correspond with the reconstructed lacustrine phases and, combined with data from Tswaing Crater, suggest that tropical moisture sources also had significant influence, at least during MIS 3 (Fig. 2). In contrast, paleolake reconstructions from the Makgadikgadi Basin of the Kalahari show limited correspondence with lacustrine phases of the South African interior (21), implying distinct responses to global drivers of climate change, cautioning against the extrapolation of northern Kalahari data to define regional climate dynamics (e.g., ref. 11).

While evidence for wetter last glacial conditions was reported as early as 1973 (24, 39), a paucity of data and minimal chronology led to generalized characterizations of the South African interior that equated lower temperatures with aridity (e.g., ref. 3). In contrast, the MIS 3-2 climatic changes implied here would have fundamentally transformed the South African central interior, rejuvenating regional hydrologic networks and significantly changing regional vegetation and faunal resources. Using a modern analog approach based on the climatic conditions inferred from the hydrological modeling, and the modern distributions of vegetation types (45) and large ungulate taxa (46), our hydroecological models imply that during lacustrine phases regional environments would have been much grassier. This vegetation would have been akin to the modern Highveld and Drakensberg Grassland bioregions, or, if increases in winter rainfall were considerable, akin to a high grass-cover shrubland (Fig. 3; *SI Appendix*, p. 61). A critical difference between modern bioregions and their MIS 2-3 climatic correlates is the edaphic contrast between the (presently) dry interior and humid uplands. While the Mesic Highveld and Drakensberg grasslands are characterized by leached soils and less palatable sourveld grasses, soils of the interior are relatively unleached, and, with increased humidity, are likely to have supported more palatable, eutrophic grasslands for which there are no direct analogs in southern Africa today (47–49). Low temperatures would have almost certainly excluded savanna bioregions (50), but woodland taxa would have likely developed in riparian and less fire-prone niches, creating a diverse vegetation mosaic. While proxy data from the interior are extremely limited, the available pollen data from Wonderwerk and Equus Cave are consistent with MIS 2-3 conditions being cooler and wetter than the Holocene, characterized by open, grassy vegetation with occasional trees (34, 51).

Faunal data from Equus Cave support these findings, with MIS 2-3 assemblages differing markedly from those of the Holocene (40). While several taxa from the Pleistocene layers are now extinct, there is no precise modern analog for the Equus Cave faunal assemblages (46). Best-fit modern assemblages for the MIS 2-3 herbivores are found in the Central Bushveld Savanna and Mesic Highveld Grasslands to the east (Fig. 3; *SI Appendix*, p. 67) (46, 47, 52). Today, these regions are substantially more humid than the Equus Cave region (aridity index (AI): 0.51 ± 0.16 compared to 0.28 ± 0.3) (53), and accordingly support a much greater herbivore biomass (10,900 kg/km^2 compared to 5,700 kg/km^2) (46). Most significant is the MIS 2-3 presence of water-dependent taxa such as hippopotamus, crocodile, and lechwe at Equus Cave (40). The latter's (*Kobus leche*) distribution is tightly linked to extensive swamps and wetlands, with the nearest modern occurrence being the Okavango Delta 900 km to the north (*SI Appendix*, p. 68). This again implies a substantial change in regional environments, with surface water occurrences not just along perennial river courses, but across the array of now-dry basins that characterize the landscape (Fig. 3), consistent with our hydrologic model and environmental reconstructions. These findings may be extended to other sites, such as Florisbad, where, despite poor chronologic control, hominin remains are associated with evidence for cool, moist grassy conditions (54) and similarly distinct fauna, including hippopotamus and lechwe (55).

Together, these results imply that during portions of MIS 3 and 2, the South African central interior experienced hydrological conditions that contrast markedly with perceptions of a persistently arid and inhospitable hinterland (Fig. 2). Instead, the region appears to have enjoyed protracted phases of abundant surface water and subsistence biomass. This allows us to account for some long-standing observations regarding the region's faunal and vegetation records, which indicate a more diverse fauna occupying a grassier landscape (34, 51). That these changes are not correlated with palaeolake phases in the northern Kalahari (21) is particularly interesting as it further highlights the spatial diversity of environmental responses to broader changes in global boundary conditions.

These findings also speak to long-standing questions regarding Pleistocene climatic conditions and human occupation of the South African interior (2–4, 11, 56–59), particularly during MIS 3 and 2. Both MSA and LSA archaeological materials occur throughout our study region and more widely across the Karoo

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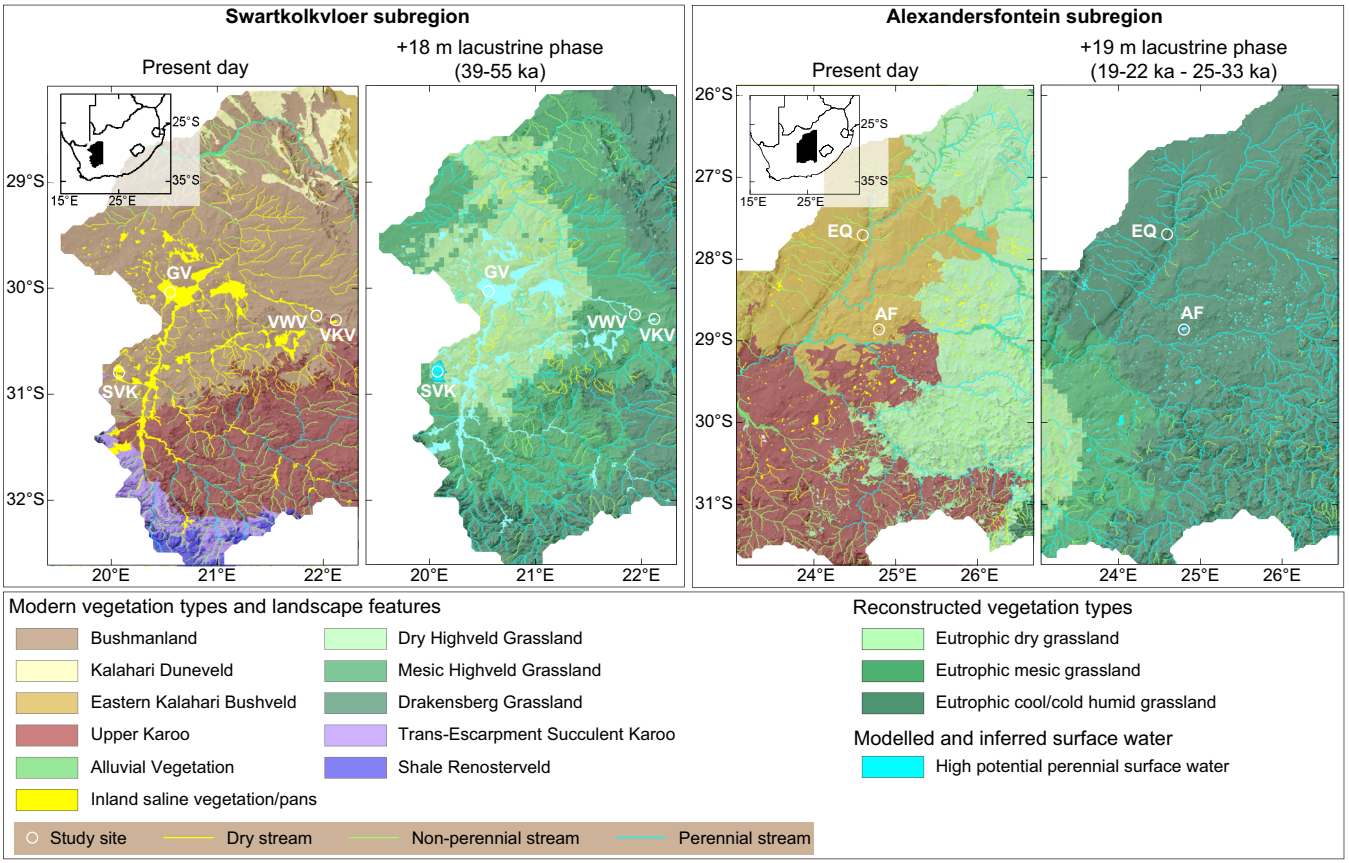


Fig. 3. The present-day conditions and environmental reconstructions for lacustrine phases. Maps of western and eastern subregions of the Orange and Vaal river catchments of the South African interior depicting the present-day hydrologic networks and vegetation bioregions as compared to hydrologies and vegetation types likely associated with key lacustrine phases identified in the geological record.

(e.g., refs. 15, 58, and 60). Often this is surficial and unstratified archaeology, but its ubiquity on and near the landforms we investigated across our broader study region (*SI Appendix*) supports Butzer's (61) contention that evidence for paleolake high-stands associates closely with diagnostically MSA artifacts. Crucially, however, we demonstrate that the most prominent waterbodies formed not in MIS 5, as previously suspected by Butzer (61), but during MIS 3-2, a timeframe broadly consistent with the character of most lake-associated lithic scatters (*SI Appendix*). It follows that the region's underrepresented archaeological record is more likely to be a product of preservation biases related to regional geologies and geomorphologies (a lack of rockshelters and deflated landscapes, respectively) than of limited resources for hunter-gatherers, particularly given the profound behavioral plasticity of Pleistocene *H. sapiens* (62, 63). While the dearth of interior rockshelters precludes assessments of demographic trends, available dating for archaeological contexts in the region implies a recurrent human presence, with extensive MSA material at sites for which our hydrological modeling clearly predicts MIS 3/2 inundation.

Southern Africa during MIS 3 and 2 witnessed major cultural changes, including the piecemeal abandonment of MSA technological traditions (64), the widespread adoption of miniaturized tools and standardized LSA microliths (65), and the establishment of new modes and magnitudes of social networking (66). Many of these changes were likely linked to environmental trends via shifting resource potentialities and demographic responses. That the central interior enjoyed, at least at times, high effective rainfall, water availability, and large ungulate-supporting grasslands enhances the likelihood that societies residing there played a role in these processes. Our findings also support evidence for potentially analogous ecological responses during earlier glacial stages—particularly MIS 6 (67), when expanded sea-ice extent coincided similarly with high austral summer insolation—and models invoking an inhospitable interior as a pretext for the existence of coastal refugia for early *H. sapiens* (e.g., refs. 6–8) may require reevaluation. For now, these findings join emerging insights from elsewhere in the interior (e.g., refs. 10–13, 16, 68, and 69) in highlighting the need to work toward a more nuanced picture of spatiotemporal variability in climatic conditions and resource availability for middle and late Pleistocene foragers. To this end, future work in the central interior should aim to integrate hydrological data like those presented here with the region's rich archaeological record by identifying and taking advantage of opportunities to chronologically constrain the latter. Only then can we appreciate when, how, and the extent to which ancient societies took advantage of the considerable opportunities afforded by glacial-phase landscapes in the subcontinent's interior basins.

Materials and Methods

Elevations for sites and analyses were established using the Multi-Error-Removed Improved-Terrain digital elevation model (70).

Geochronology (*SI Appendix*, p. 3). Samples of the lacustrine gastropod *T. ventricosa* (Swartkolkvloer) and the freshwater bivalve *U. caffer* (Grootvloer) were manually extracted from encapsulating sediments under a binocular microscope. They were vigorously cleaned in deionized water with an ultrasound bath and inspected to ensure no adhering sediment or cement were present. Where possible, whole shells of *T. ventricosa* were analyzed. Samples were submitted to the ^{14}C Chrono Centre at Queen's University Belfast (samples UB-) or to Beta

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Analytical (samples Beta-). Radiocarbon ages were calibrated using CALIB 8.1.0 (71) using the SHCal20 Southern Hemisphere calibration dataset (72). As it presently remains unquantified, no reservoir effect corrections were made. The results are summarized in *SI Appendix*, *Appendix A*.

Luminescence dating samples were obtained by hammering steel tubes into cleaned exposures or by using a Dormer sand augering system. Samples were prepared for coarse grain (range 90 to 250 μm) quartz and/or K-feldspar analysis following standard methods (73). Luminescence measurements on (2 mm) multigrain aliquots were performed on a Risoe DA20 TL/OSL reader. For quartz, stimulation was provided by blue LEDs (stimulation wavelength 470 nm) at 70% power (40 s at 125 °C) with the resulting luminescence detected through a U-340 detection filter. For K-feldspar, stimulation was provided by IR diodes (wavelength 870 nm) with detection through Schott BG39 filters and Corning 7-59 filters (detection range 320 to 450 nm). Equivalent doses (D_e) were determined using mineral-specific variants of the single-aliquot regenerative-dose (SAR) protocol (74, 75). These comprised 5 to 7 regeneration dose point sequences and included a repeat dose point and a zero-dose point. For K-feldspar, the post-IR IRSL method (76–78) was used. This comprised an initial stimulation at 50 °C followed by a second (post-IR) stimulation at 225 °C ($pIRIR_{225}$) or 290 °C ($pIRIR_{290}$). K-feldspar fading rates were measured using subsets of aliquots from the SAR measurements following Auclair et al. (79), and fading rates (g-$value_{2days}$) and any fading corrections were analyzed in the "*Luminescence*" R Package (80). For both minerals, aliquots with recycling ratios (repeat dose points in the dose–response curve) outside 10% of unity, or recuperation (sensitivity-corrected zero dose L_x/T_x) >5% of the sensitivity-corrected natural signal (L_n/T_n), were removed prior to averaging of the D_e distributions. The dose-response curves were fitted with saturating exponential fits, with D_e uncertainties incorporating counting statistics, curve-fitting uncertainties, and a 1.5% systematic uncertainty [calculated with the Risoe *Analyst* software (81)]. The uncertainty in the final D_e estimates incorporates a beta source calibration uncertainty (3%).

Single-grain quartz analyses employed a focused 532 nm Nd:YVO_4 solid-state diode-pumped laser emitting at 532 nm, focused to a ~20 μm spot. The dose-response curves were fitted with saturating exponential fits in the *Analyst* software and their uncertainties included the addition of a 3% systematic instrumental uncertainty (determined following Jacobs et al. 82). Grains were rejected (e.g., ref. 83) if: 1) the test dose signal was less than 3 sigma above background; 2) the relative test dose uncertainty was larger than 20%; 3) the recycling ratio or IR depletion ratios were inconsistent with unity at two sigma; 4) recuperation (zero dose L_x/T_x) was greater than 5% of L_n/T_n; 5) the dose-response curve did not grow with dose; and 6) a finite D_e or D_e uncertainty could not be obtained due to saturation of the dose-response curve, or a failure of the L_n/T_n measurement to intercept the dose-response curve. The results are summarized in *SI Appendix*, *Appendix A*.

Environmental dose rates were determined using elemental concentrations obtained via inductively coupled plasma-mass spectrometry (ICP-MS; for U and Th) and ICP-OES (for K). External beta dose rates were further evaluated using GM beta counting (84). For some samples (*SI Appendix*), in situ gamma spectrometry measurements were undertaken (85). U, Th, and K concentrations were converted to annual dose rates following Guerin et al. (86), then corrected for grain size (87), water content (85), and HF etching (88). Water contents were obtained using the modern measured water contents in most cases, but adjusted when evidence of modern drying of exposed sections was readily apparent (see *SI Appendix* p. 5). For K-feldspar, internal grain K and Rb concentrations of 12.5 ± 0.5% and 400 ± 100 ppm, respectively, were assumed (89, 90). HF etching was assumed to have removed the α-irradiated outer portion of the quartz and (where used) feldspar grains. For the un-etched K feldspar samples, an external alpha dose rate used an a-value of 0.15 ± 0.05 (91). Cosmic dose rates based were determined using the "*Luminescence*" R package ((80), and *SI Appendix*) based on measured sample depths.

Hydrological Modeling (*SI Appendix*, p. 56). The Shetran hydrological model (38, 92) is physically based and spatially distributed and has a full three-dimensional subsurface flow component. In Shetran, there is an explicit representation of the groundwater and surface flows around and within the pan sites, including components for vegetation interception and transpiration, overland flow, variably saturated subsurface flow, and channel–aquifer interactions. Solutions to the governing physics-based partial differential equations of mass and momentum are achieved on a three-dimensional grid using finite-difference equations (https://research.ncl.ac.uk/shetran/). The Orange River catchment was split into 28,591 4 km × 4 km vertical columns. Each vertical column has an associated vegetation type on the ground surface and below ground is split into 35 cells, to 80 m below the ground surface, with small cells near the surface containing the soils and larger cells deeper below ground containing the aquifer. In addition, there are 1,165 river channels that flow around the edge of the grid squares, automatically generated from the ground surface elevations (38).

SI Appendix, Table S8 shows the original sources of spatial distributed data used by Shetran, which were aggregated up to a common 4 km resolution. The northwestern part of the basin is classed as being bare ground, the central part as shrubland, and the eastern part as a mixture of arable and grassland. The bare ground corresponds to "Bushmanland," shrubland to "Upper Karoo" and "Eastern Kalahari Bushveld" and arable and grassland to "Dry Highveld Grassland", "Mesic Highveld Grassland," and "Drakensberg Grassland" (see *Vegetation Reconstruction* section, *SI Appendix*, Fig. S59). Soils (*SI Appendix*, Table S8) are generally well drained and either clay loams, sandy clay loams, sandy loams, or sandy. Standard libraries were used for the soil and land-use parameter values (93). There is limited information about the aquifer properties within the catchment. As such, the same aquifer properties were applied throughout the model. The average spatially distributed monthly values of precipitation and potential evaporation were obtained from the WorldClim 2.1 dataset (94) and Global AI and Potential Evapotranspiration (ET0) Climate Database v2 (53). There is an increase in annual totals of precipitation going from west (<200 mm a^{-1}) to east (>1,000 mm a^{-1}). Potential evaporation totals show a decrease from west (>1,600 mm a^{-1}) to east (<1,000 mm a^{-1}).

The aim of the hydrological simulation (outputs shown in *SI Appendix*, Figs. S51 and S53 and Table S9) was to produce the expected long-term water table and surface water depths, as observed and preserved at the lake basin study sites. The model was run for up to 100 y with the same annual precipitation and potential evaporation data until equilibrium conditions were achieved. Evapotranspiration is affected by the temporal distribution of the precipitation, so the monthly precipitation data were converted to daily precipitation using the expected number of wet days per month (https://www.fao.org/aquastat/en/geospatial-information/climate-information).

Palaeoecological Analysis (*SI Appendix*, pp. 61, 67). To consider changes to vegetation and large ungulate resources associated with the inferred MIS 3-2 lacustrine periods, we employed a modern analog approach, based on the modern/historic distribution of biomes and bioregions (45) and fauna (46) and comparable climate data (53, 94).

The distribution of biomes in the interior is primarily determined by: 1) water availability (e.g., discriminating between Grassland and Nama-Karoo; *SI Appendix*, Fig. S54) and 2) mean minimum temperature of the coldest month (TminColdM; a proxy for frost potential, e.g., discriminating between Grassland and Savanna; *SI Appendix*, Fig. S54*D*) (50, 95). At the subcontinental scale, the seasonal distribution of rainfall is also significant, discriminating the winter-rainfall biomes of the Fynbos and Succulent Karoo of the southwestern Cape from the summer-rainfall bioregions of the rest of southern Africa (*SI Appendix*, Fig. S54*C*). Rainfall seasonality cannot be accurately quantified from the available proxy records, but our conclusions encompass the spectrum of vegetation types possible under the summer and winter-rainfall-dominant scenarios. To estimate palaeotemperature during the lacustrine phases, we considered the proxy-based palaeotemperature data available from southern Africa, which typically indicate a temperature depression of ~4 to 7 °C at the Last Glacial Maximum (19 to 26.5 ka) (96), and an overall pattern that tracks global trends, as characterized by Antarctic ice core data (97). We used these patterns and data to create a temperature time series for the region, calibrated using regional temperature estimates (for amplitude of change) and the EPICA Antarctic temperature reconstruction (98) (for the timing of change) (97) (*SI Appendix*, Fig. S56). Using these data, we established a temperature-based control on the potential influence of PET on the water balance during lacustrine phases, and then adjusted mean annual precipitation values to achieve the target P:PET ratios indicated by the Shetran hydrological model (*SI Appendix*, Table S9). The MAP and PET data were used to calculate spatially continuous AI grids, the input values of which could be modified to reconstruct the water balance for the lacustrine phases in the South African interior.

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To consider changes in regional fauna associated with MIS 2 to 3 environments, we reanalyzed data from Equus Cave, the only faunal record from the South African interior that preserves remains dated to MIS 1, 2, and 3. The nature of the site's chronology does not permit detailed time series analyses, but it is sufficient to assess late Pleistocene assemblages (40, 99). For our analysis, we compared the large ungulates reported by Klein et al. (40) on a presence/absence basis with comparable modern/historic data compiled by Hempson et al. (46) for sub-Saharan Africa (*SI Appendix*, Fig. S60).

Data, Materials, and Software Availability. All study data are included in the article and/or *SI Appendix*.

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